

Keshav Rajput

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Software engineer, new grad. MS Computer Science, NYU Tandon, May 2026. C, C++, Python, TypeScript, Kotlin.

github.com/happyc0der | linkedin.com/in/keshav-rajp4-862267259 | happyc0der.github.io

EDUCATION

New York University, Tandon School of Engineering

Sep 2024 - May 2026

Master of Science (MS) in Computer Science

GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi

2020 - 2024

Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics

CGPA 8.19/10

Coursework: Algorithms II, Database Systems, Open Source and Professional Software Development, Distributed Systems, Operating Systems, Computer Networks

PROJECTS

Atlas (in progress) | C++23, SDL_GPU, HLSL | github.com/happyc0der/atlas-engine

2026

- Solo C++23 engine for map-based strategy games, 25 gated milestones: SDL_GPU renderer, hot-reload assets, versioned serialization, 21 ADRs.
- Deterministic fixed-tick simulation with state hashing, replay and lockstep over ENet, proved hash for hash across two processes; million-cell tick in 694 us.
- Mods run as WebAssembly with no clock, filesystem or RNG; 1045 tests, CI on macOS, Linux and Windows with ASan/TSan and gated benchmarks.

mlds-auth | C11, liboqs, libsodium, SQLite, CMake | github.com/happyc0der/mlds-auth

2026

- Post-quantum mutual authentication protocol in C11: ML-DSA-65 (FIPS 204) signatures over hybrid X25519 + ML-KEM-768, HKDF-SHA256, ChaCha20-Poly1305 records.
- 0.45 ms median handshake, 736 MiB/s record sealing; ASan/UBSan CI on Linux and macOS, 12 libFuzzer targets, mutation testing with 192 must-kill mutations nightly.
- Built mlds-authd on it: login codes, device enrolment, revocation, key rotation, recovery codes; ProVerif model of the handshake.

doom-ports | C, eZ80 assembly, Monkey C, Connect IQ SDK | github.com/happyc0der/doom-ports

2026

- From-scratch first-person raycaster for the TI-84 Plus CE in C and eZ80 assembly: 24 KB, 5 to 7 fps, all art generated at startup.
- Measured 122 eZ80 cycles per VRAM byte, fit a frame-time model (ceiling 6.2 fps) and moved the column fill to hand-written assembly.
- Garmin Venu X1 port in Monkey C, rebuilt around a 0.2 ms per-draw cost: 1 to 2 fps to 15.6 fps; on the Connect IQ store.

TideFit | TypeScript, Next.js 16, React 19, Supabase | github.com/UtkarshMitta/tide-fit

2026

App by Utkarsh Mittal; the audit, upgrade, tests, CI and launch are mine.

- Security audit (code review of all 45 source files) of a Next.js trip planner: 15 vulnerabilities confirmed on the running app (OAuth callbacks accepting any authorization code, a row-level security policy exposing every user, a plaintext secret cookie), 13 fixed.
- Upgraded Next 14 to 16 and React 18 to 19 by hand across 16 merged pull requests; npm audit 6 vulnerabilities to 0, tests 0 to 45, CI (typecheck, lint, tests) deploying main to Vercel.
- Launched at tide-fit.vercel.app: private Blob storage, edge rate limit verified at 20 requests then 429, Strava and Google sign-in verified live.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **Gemma 4 Developer Agent, Kaggle (Google DeepMind)**, 2026. LLM agent entry in Google ADK; local evaluation harness with Docker sandboxes and tool calls, served by vLLM. In progress, ends 2026-12-02.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Systems: CMake, gRPC, PostgreSQL, SQLite, WebAssembly (sandboxed mods), Android (Jetpack Compose, Room), Next.js, React, FastAPI, Node.js, REST APIs, MCP

Testing: pytest, Catch2, vitest, Robolectric, libFuzzer, ASan/UBSan, static analysis (clang-tidy, ruff, ESLint), ProVerif, unit and integration tests, CI

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes with regression tests to hawk-py
- Treasurer, OWASP Student Chapter, IIIT Delhi